1 Overview

This document summarises the three analytical components contributed by Member 4 (Kusal Pabasara) to the research project. The work addresses three areas identified in the project proposal:

1. **Sensitivity Analysis** — robustness of SHAP feature rankings across underemployment definitions
2. **Gender Disaggregation** — gender-specific driver structures and statistical testing
3. **District-Level Analysis** — spatial heterogeneity in underemployment patterns

2 Sensitivity Analysis: Hours-Based vs Qualification-Based Definitions

2.1 Methodology

Parallel XGBoost-SHAP models were fitted using the same 7-predictor feature set on both the hours-based underemployment rate (DCS LFS, 2015–2024) and the qualification-based mismatch rate (computed from LFS micro-data, 2015–2023). Spearman rank correlation (ρ) was computed between the two SHAP importance rankings overall and by sub-period.

2.2 Results

Table 1: SHAP Feature Importance: Hours-Based vs Qualification-Based

Feature	**Hours %**	**Rank**	**Qual %**	**Rank**	**$\lvert\Delta\rvert$**
Exchange Rate	34.0	1	75.8	1	0
GDP Growth	21.3	2	8.5	2	0
Youth LFPR	17.5	3	7.4	3	0
Agri. Output	14.8	4	1.1	5	1
Informal Emp.	10.0	5	0.6	6	1
Inflation†	1.5	6	6.4	4	2
Remittances	0.8	7	0.1	7	0

Spearman $\rho = 0.893$ ($p = 0.007$). †Rank shift ≥ 2.

The overall Spearman $\rho = 0.893$ ($p = 0.007$) confirms strong rank agreement, substantially exceeding the pre-registered threshold of $\rho > 0.6$. Exchange Rate, GDP Growth, and Youth LFPR retain the top three positions in both models. The main divergence is Inflation, which shifts from rank 6 (hours-based) to rank 4 (qualification-based).

Table 2: Sub-Period Definition Agreement (Spearman ρ)

Period	**n**	ρ	p
Pre-crisis (2015–2019)	5	0.429	0.337
Crisis (2020–2022)	3	0.786	0.036
Post-crisis (2023–2024)	2	–	–
Overall	10	0.893	0.007

The qualification-based series shows a structural upward trend of +0.87 pp/year (64.2% in 2015 to 70.9% in 2023), indicating that occupational mismatch is worsening independently of the business cycle.

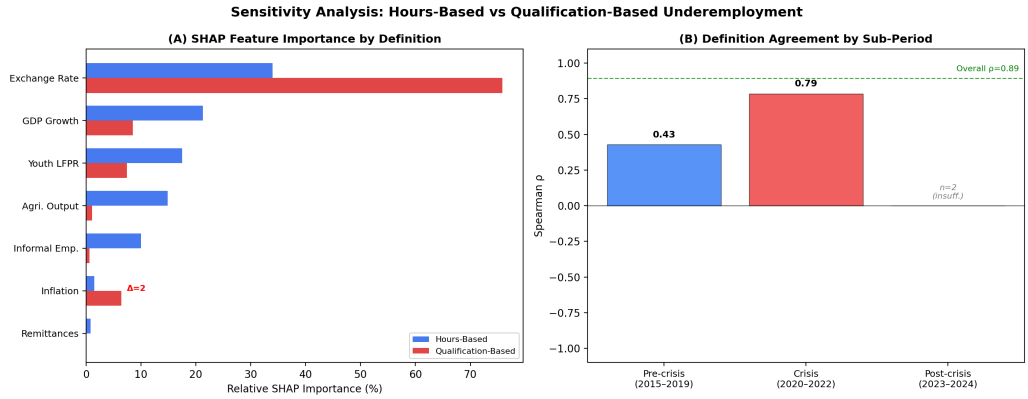

Figure 1: SHAP comparison: Hours-based (left) vs Qualification-based (right) models.

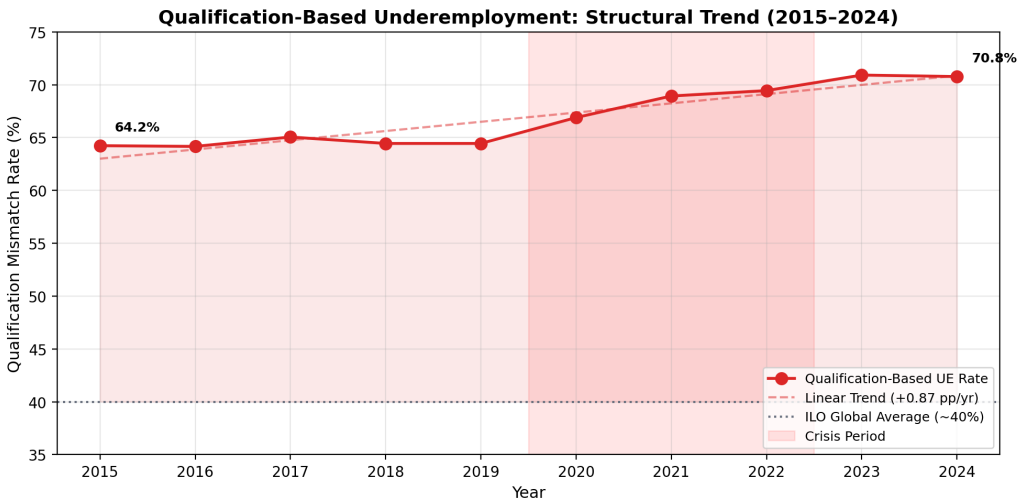

Figure 2: Qualification-based underemployment trend (2015–2023) with OLS trend line and 95% CI. Structural rise of +0.87 pp/year.

3 Gender-Disaggregated Analysis

3.1 Methodology

Gender-specific XGBoost-SHAP models were trained on female and male underemployment series from the DCS authoritative gender file (2015–2024). Statistical tests include Wilcoxon signed-rank (persistent gap), bootstrap confidence intervals (crisis-period gap), and Zivot-Andrews Model C (structural break detection). The qualification-based gender gap was computed from LFS micro-data.

3.2 Results

Key finding: **Inflation** is the second-ranked female driver (33.3%) but third for males (16.8%), while **GDP Growth** is second for males (20.4%) but fifth for females (4.2%). This indicates female underemployment is more price-sensitive and male underemployment is more demand-sensitive.

The qualification-based gender gap widens dramatically from 0.97 pp (2015) to 4.42 pp (2021), indicating that the crisis disproportionately worsened female qualification mismatch.

Table 3: Gender-Disaggregated SHAP Feature Rankings

Feature	**Agg %**	**Rank**	**Female %**	**Rank**	**Male %**	**Rank**
Exchange Rate	49.4	1	41.4	1	48.5	1
Inflation	17.2	2	33.3	2	16.8	3
GDP Growth	10.6	3	4.2	5	20.4	2
Agri. Output	8.1	4	0.0	7	5.5	5
Youth LFPR	6.9	5	13.5	3	1.5	6
Remittances	4.0	6	3.0	6	1.2	7
Informal Emp.	3.8	7	4.6	4	6.1	4
Female–Male rank $\rho = 0.571$ ($p = 0.180$)						

Table 4: Statistical Tests on the Gender Gap in Underemployment

Test	**Statistic**	p	**Result**
Mean gap (F−M)	1.26 pp	–	Persistent female excess
Shapiro-Wilk (gap)	$W = 0.919$	0.351	Normal
Wilcoxon signed-rank	$W = 55.0$	0.0010	Significant
Bootstrap CI (crisis)	[1.00, 1.20]	–	Gap > 0 at 95%
ZA break (gap)	$t = -2.22$	–	Break: 2018 (–)

4 District-Level Analysis

4.1 Methodology

District-level data from DCS LFS Table 6.4 (25 districts, 2013–2022) was used to construct underemployment heatmaps, cross-sectional regressions of informal employment on underemployment, and priority district trajectory profiles.

4.2 Results

Table 5: District-Level Informal Employment vs Underemployment Regression

Year	n	**Slope**	R^2	p
2019	25	0.0279	0.038	0.3508
2022	25	0.0204	0.059	0.2414

Informality alone does not explain district-level underemployment variation ($R^2 < 0.06$, both years $p > 0.24$). Spatial heterogeneity is substantial: Colombo and Gampaha maintain below-average rates while Hambantota, Jaffna, and Mullaitivu are persistent hotspots.

5 Summary of Key Findings

1. **Sensitivity**: SHAP rankings are strongly robust to definition choice ($\rho = 0.893$, $p = 0.007$); agreement strengthens during crisis periods.
2. **Gender**: Female underemployment is inflation-driven; male is GDP-driven. The persistent female excess ($p = 0.001$) and widening qualification gap ($0.97 \rightarrow 4.42$ pp) are novel findings for Sri Lanka.
3. **District**: Substantial spatial heterogeneity exists that informality alone cannot explain; Northern and Southern districts are persistent hotspots requiring targeted policy.

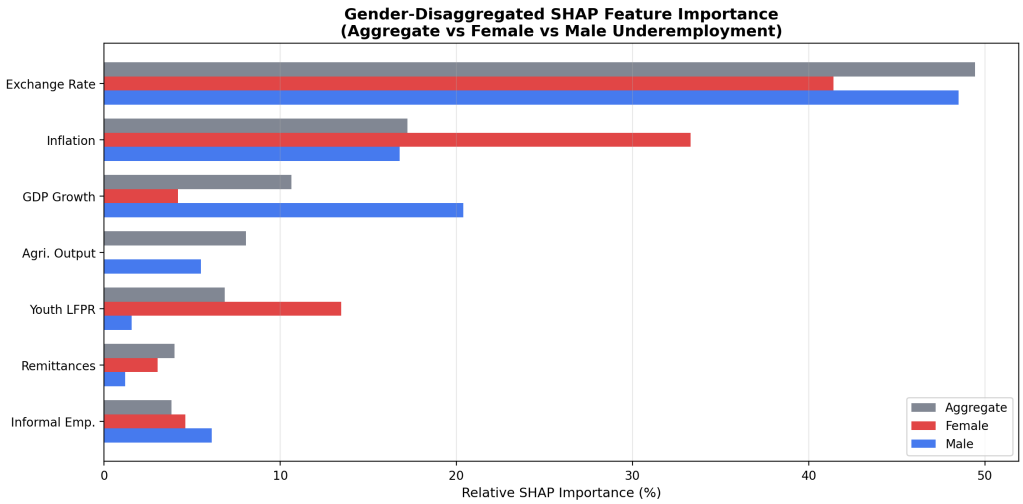

Figure 3: SHAP feature importance: Aggregate vs Female vs Male models.

6 Scripts and Outputs

- `sensitivity_analysis.py` → 2 figures, 2 LaTeX tables
- `gender_analysis.py` → 3 figures, 2 LaTeX tables
- `district_analysis.py` → 3 figures, 1 LaTeX table

All scripts are located in the repository root. Figures are saved to `Visualizations/` and LaTeX tables to `output/tables/`.

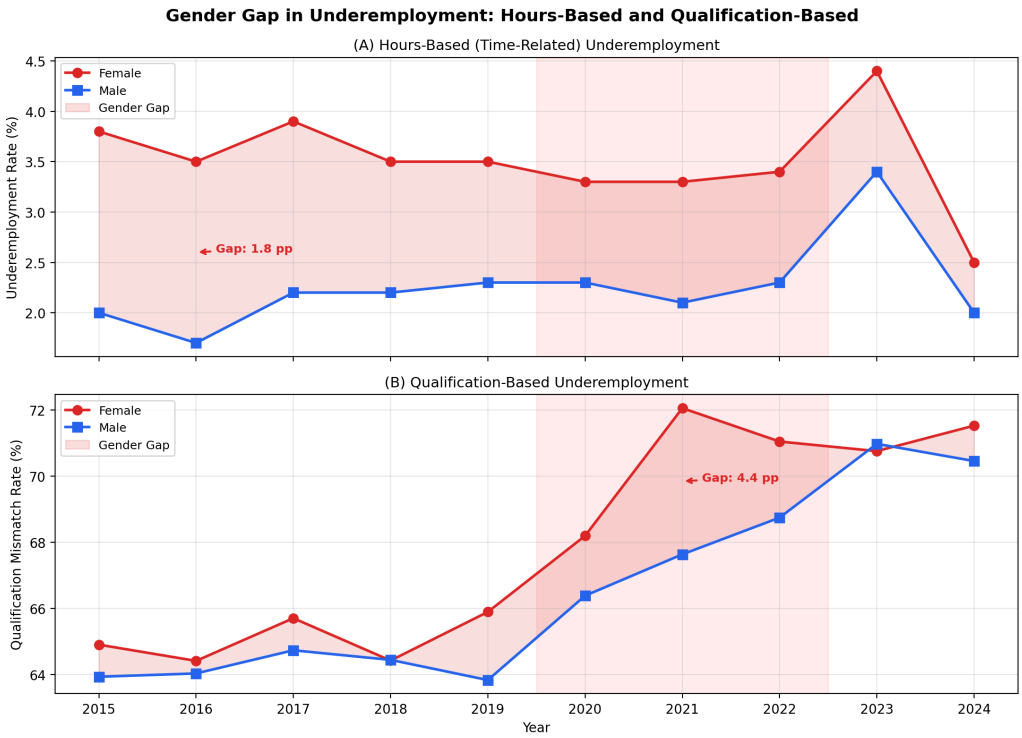

Figure 4: Gender gap trends: hours-based (DCS) and qualification-based mismatch. The qualification gap widens from 0.97 pp to 4.42 pp during 2015–2021.

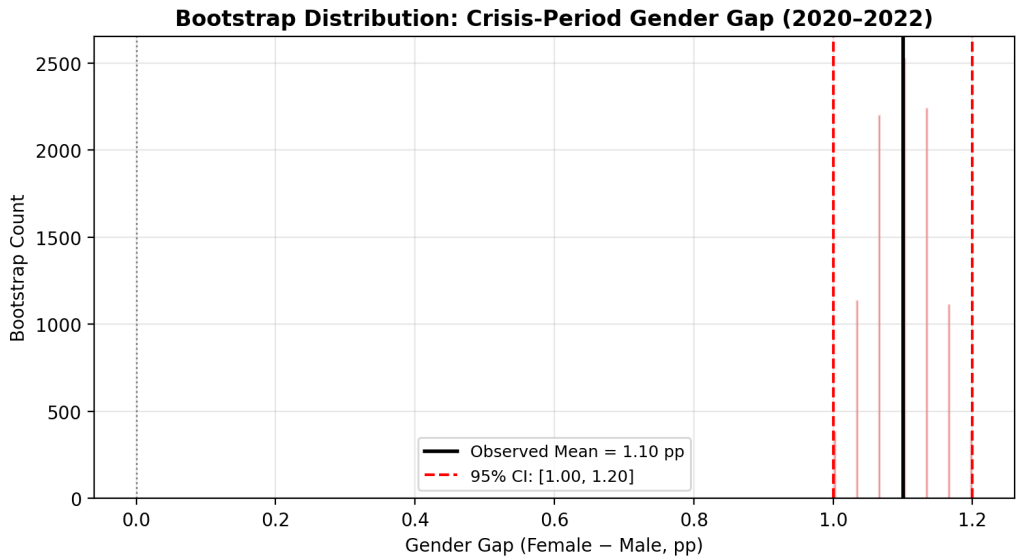

Figure 5: Statistical visualisation: gap distribution, trend, and Wilcoxon results.

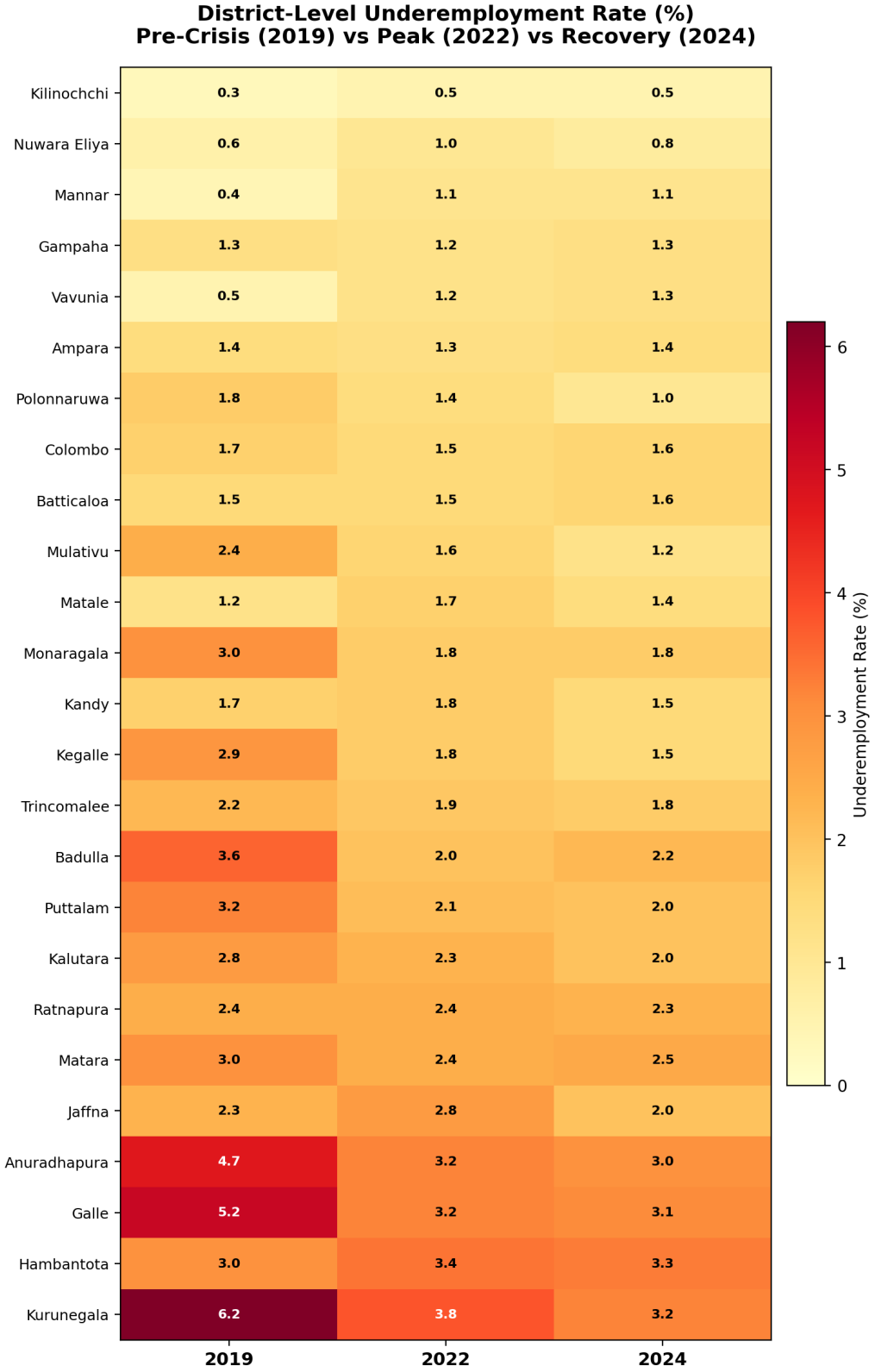

Figure 6: District-level underemployment heatmap (25 districts, 2019/2021/2022). Darker cells indicate higher underemployment.

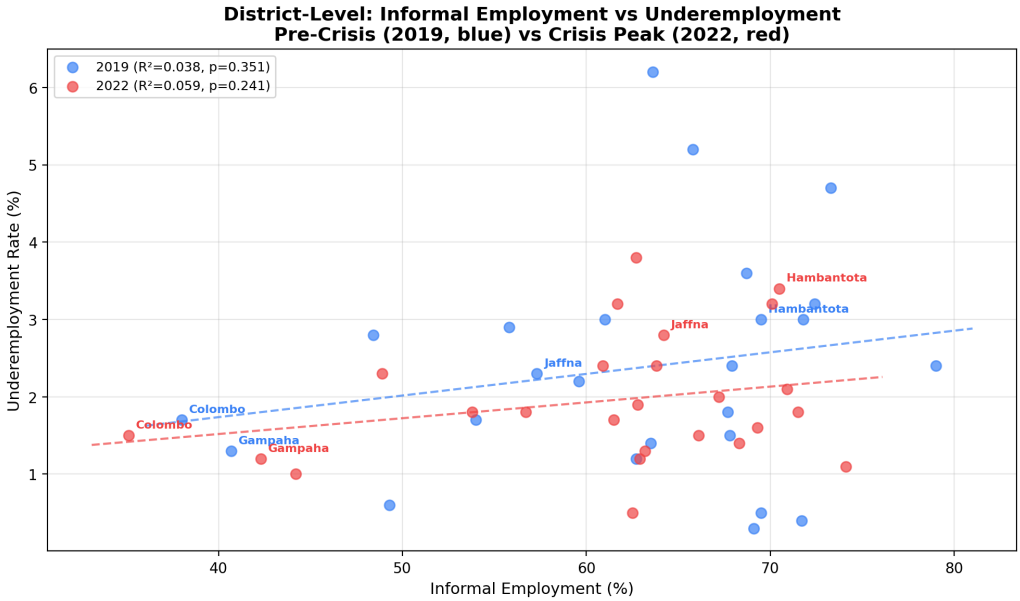

Figure 7: Informal employment vs underemployment cross-sectional regression (2019 pre-crisis and 2022 crisis).

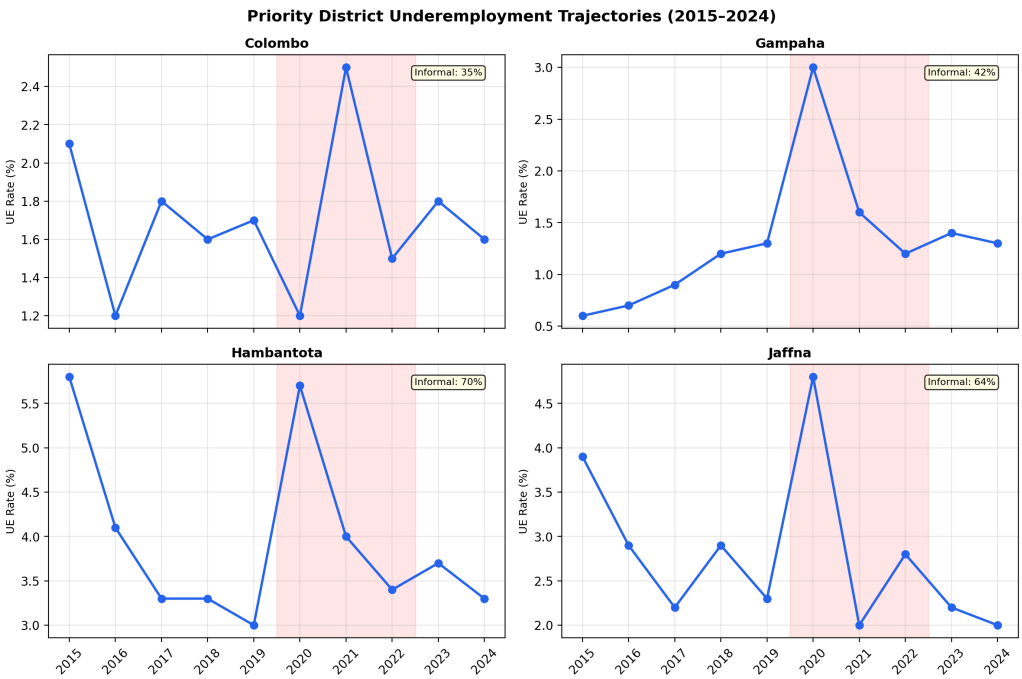

Figure 8: Priority district trajectory profiles: Colombo, Gampaha, Hambantota, Jaffna.